

Building 101/102: Re-Heat Exchanger Sequence of Operation

Run Conditions - Requested:

The re-heat water shall be run whenever:

- Any main AHU (S1 through S3) is indexed on
- OR radiant heating is indexed on.
- OR the re-heat heating is overridden on.

Re-Heat Heating Pump:

The re-heat pump shall be energized when the re-heat system is commanded to run. To prevent short cycling, the pump shall have a minimum runtime of 10 minutes (adj.). A pressure differential switch shall monitor the pump status.

Alarms shall be provided as follows:

- Pump Failure or switched "Off" locally: Commanded on, but the status is off.
- Pump switched "On" locally: Commanded off, but the status is on.

To ensure residual heat is carried away from the re-heat heat exchanger when the re-heat heating is commanded off, the high temperature control valve shall be commanded shut and the re-heat hot water pump shall remain running for 10 minutes (adj.).

Re-Heat Heating Water Temperature:

The controller shall monitor the re-heat heating pump status and enable the high temperature hot water controls when the pump is detected to be operating.

The high temperature hot water control valve shall not be enabled until the re-heat heating pump status indicated the pump has been running for 5 minutes (adj.) to ensure substantial water flow is established.

The controller shall monitor the re-heat circuit supply temperature and modulate the high temperature hot water valve operation to maintain a hot water supply temperature setpoint as scheduled against the outside air temperature, as follows:

Outside Temp:	60°F (adj.)	10°F (adj.)
Circuit Temp:	140°F (adj.)	190°F (adj.)

Alarms shall be provided as follows:

- If the re-heat circuit supply temperature is greater than +10°F (adj.) above the scheduled setpoint for 10 minutes (adj.).
- If the re-heat circuit supply temperature is less than -20°F (adj.) below the scheduled setpoint for 10 minutes (adj.) and after the system has been operating for 60 minutes (adj.).

END OF SEQUENCE